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Centre Number

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Student Number

2021 YEAR 12

Advanced Mathematics

Task 1 Validation Task

9/12/21

Q	Marks
1	
2	
3	
4	
5	
6	
7	
8	
9	
Total	

General

Instructions:

- Reading time – 5 minutes
- Working time – 35 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations
- No white-out may be used

Total Marks: 28

This question paper must not be removed from the examination room.

This Validation task constitutes 60% of the Assignment.

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In Questions 1 – 9, your response should include relevant mathematical reasoning and/or calculations.

Question 1 (2 marks)

Solve $3^x = 25$, giving your answer to 2 significant figures.

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Question 2 (2 marks)

Solve $3 \log_2(10) + \log_2(x) = 6$. Express answer in simplest form.

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Question 3 (3 marks)

Solve $2^{x^2} = 8^{\frac{2}{3}x+1}$.

3

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Question 4 (3 marks)

Differentiate the following, simplifying where necessary.

a) $y = 2e^{4x-3}$

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b) $y = (2 - e^{-x^2})^6$

2

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Question 5 (2 marks)

If $\log_x(6) = a$ and $\log_x(3) = b$, find $\log_x(8)$ in terms of a and b .

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Question 6 (5 marks)

Radioactive decay is a process which causes a substance to break down over time. The half-life of a substance refers to the amount of time it takes for the substance to decay to half its original mass.

Consider the formula for the radioactive decay of the substance carbon-14,

$$C = C_0 e^{-kt},$$

where C_0 and k are constants, t is time in years, and C is the amount of carbon-14 measured in grams.

- a) Find the value of C_0 if there is initially 1000 grams of carbon-14. 1

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- b) The half-life of carbon-14 is known to be 5730 years. 3

Use this to show that $k = \frac{\ln(2)}{5730}$.

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- c) Hence, find the amount of carbon-14, to the nearest gram, after 10,000 years. 1

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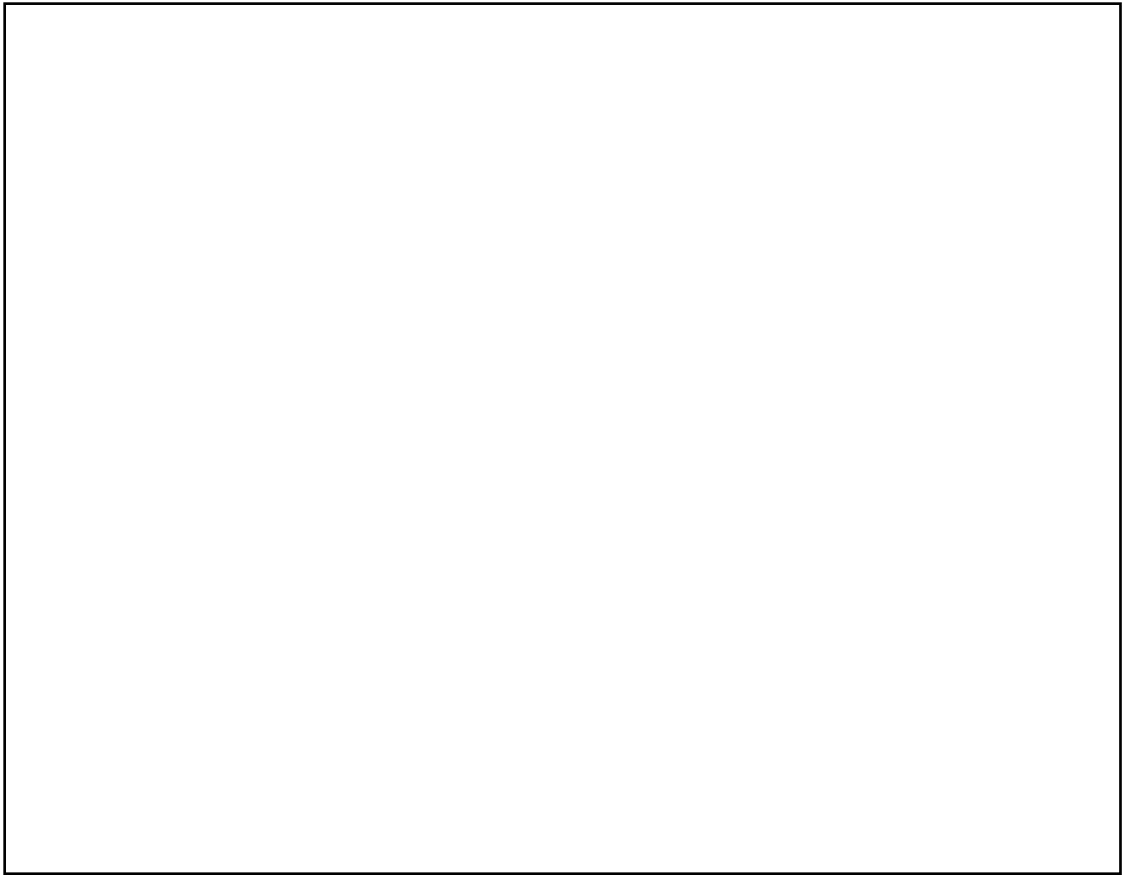
Question 7 (3 marks)

- (a) Sketch the graph of the following exponential function in the box below.
Label x and y intercepts and the asymptote.

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$$y = e^{-x} - 2$$

Note: The box suggests the size of the expected graph



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Question 8 (6 marks)

Sound intensity level (L) measured in decibels (dB) is a logarithmic measure given as:

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

The reference level of the intensity of sound (I_0) is 10^{-12} watts/m².

- (a) The table below shows the average sound intensity levels (L) and the intensity of the sound (I) of four instruments of a symphony orchestra.

2

Musical Instrument	Average sound intensity level, L (to nearest dB)	Intensity of sound, I (watts/m ²) to 7 decimal places
Cello	87	(i)
Violin	93	0.0019953
Trombone	100	0.0100000
Timpani and bass drum rolls	(ii)	0.0501187

Determine the missing values of (i) and (ii).

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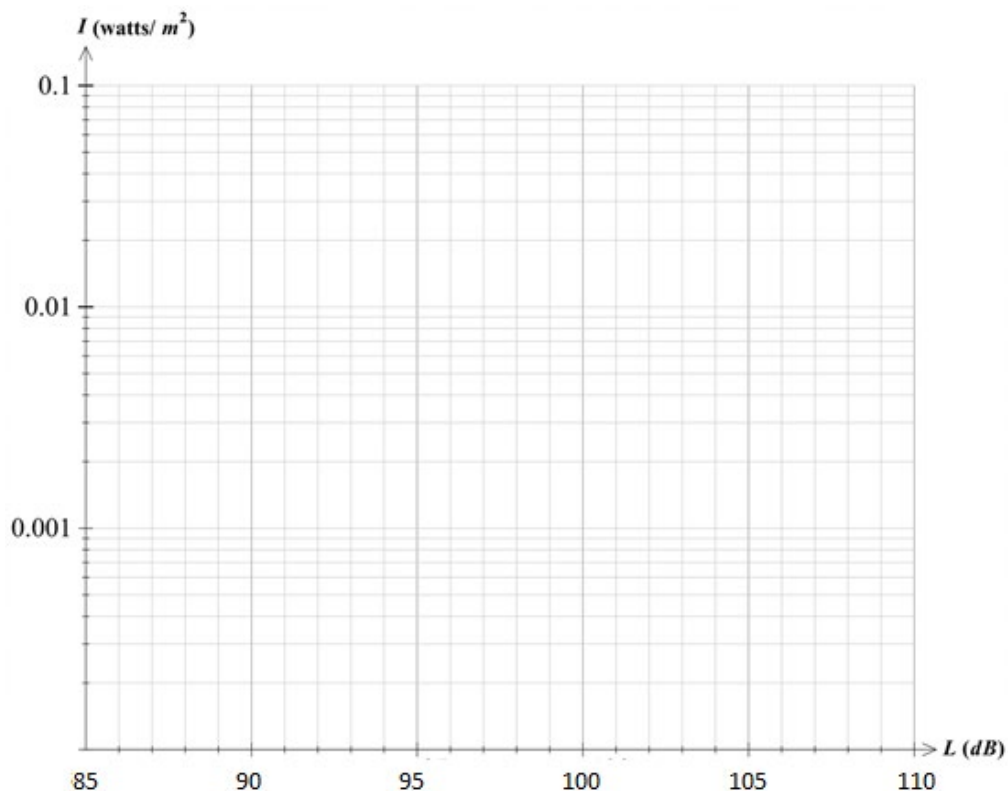
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- (b) Plot the intensity of sound (I) against the average sound intensity levels (L) for the **four instruments listed in part (a)** on the semi-logarithm graph below. **2**
Connect the points with a line.



- (c) Using the graph determine the average sound intensity level(dB) of the sound from a musical instrument that has an intensity of 0.006 watts/m^2 . **1**

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- (d) Explain why a semi-logarithm grid is useful for this data. **1**

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Question 9 (2 marks)

Given $\log_c(y) = \log_c(A) + tk \log_c(b)$, where A, b, c, k are (constant) positive real values with $b \neq 1$ and $c \neq 1$, and t and y are variables, show that $y = Ab^{kt}$.

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Examination continues on next page

Extra writing space

If you use this space, clearly indicate which question you are answering.

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In Questions 1 – 9, your response should include relevant mathematical reasoning and/or calculations.

Question 1 (2 marks)

Solve $3^x = 25$, giving your answer to 2 significant figures.

2

$$x = \log_3(25)$$

① mark

$$x = 2.9$$

① mark (2 sig. fig.)

Question 2 (2 marks)

Solve $3 \log_2(10) + \log_2(x) = 6$. Express answer in simplest form.

2

$$\log_2(10^3) + \log_2(x) = 6$$

$$\log_2(1000x) = 6$$

$$1000x = 2^6$$

$$x = \frac{2^6}{1000}$$

$$x = \frac{8}{125}$$

① mark: correctly simplifying surds

① mark: fully simplified

Question 3 (3 marks)

Solve $2^{x^2} = 8^{\frac{2}{3}x+1}$.

3

$$2^{x^2} = (2^3)^{\frac{2}{3}x+1}$$

$$2^{x^2} = 2^{2x+3}$$

$$\Rightarrow x^2 = 2x+3$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$x = 3, -1$$

① mark: correctly converting to same base

① mark: recognising quadratic

① mark: correct solutions

Examination continues on next page

Question 4 (3 marks)

Differentiate the following, simplifying where necessary.

a) $y = 2e^{4x-3}$

1

$$\frac{dy}{dx} = 8e^{4x-3}$$

① mark

b) $y = (2 - e^{-x^2})^6$

2

$$\frac{dy}{dx} = 6(2 - e^{-x^2})^5 \times 2xe^{-x^2}$$

① mark

$$= 12xe^{-x^2}(2 - e^{-x^2})^5$$

① mark: fully simplified

Question 5 (2 marks)

If $\log_x(6) = a$ and $\log_x(3) = b$, find $\log_x(8)$ in terms of a and b .

2

$$\log_x(8) = \log_x(2^3)$$

$$= 3\log_x(2)$$

$$= 3[\log_x(6) - \log_x(3)]$$

① mark: recognising 8 is a power of 2 and simplifying

$$= 3(a - b)$$

① mark: correct answer

Question 6 (5 marks)

Radioactive decay is a process which causes a substance to break down over time. The half-life of a substance refers to the amount of time it takes for the substance to decay to half its original mass.

Consider the formula for the radioactive decay of the substance carbon-14,

$$C = C_0 e^{-kt},$$

where C_0 and k are constants, t is time in years, and C is the amount of carbon-14 measured in grams.

- a) Find the value of C_0 if there is initially 1000 grams of carbon-14.

1

$$1000 = C_0 e^{-k \times 0}$$

$$C_0 = 1000$$

① mark: correct answer

- b) The half-life of carbon-14 is known to be 5730 years.

3

Use this to show that $k = \frac{\ln(2)}{5730}$.

$$500 = 1000 e^{-5730k}$$

$$e^{-5730k} = \frac{1}{2}$$

① mark: correct value of C and t

$$-5730k = \ln\left(\frac{1}{2}\right)$$

$$k = \frac{\ln\left(\frac{1}{2}\right)}{-5730}$$

① mark: correctly solving for k

$$k = \frac{\ln(2^{-1})}{-5730}$$

$$k = \frac{\ln(2)}{5730}, \text{ as req.}$$

① mark: recognising $\frac{1}{2} = 2^{-1}$ and simplifying

- c) Hence, find the amount of carbon-14, to the nearest gram, after 10,000 years.

1

$$C = 1000 \times e^{-\frac{\ln(2)}{5730} \times 10,000}$$

$$= 298g$$

① mark: answer, correctly rounding

Examination continues on next page

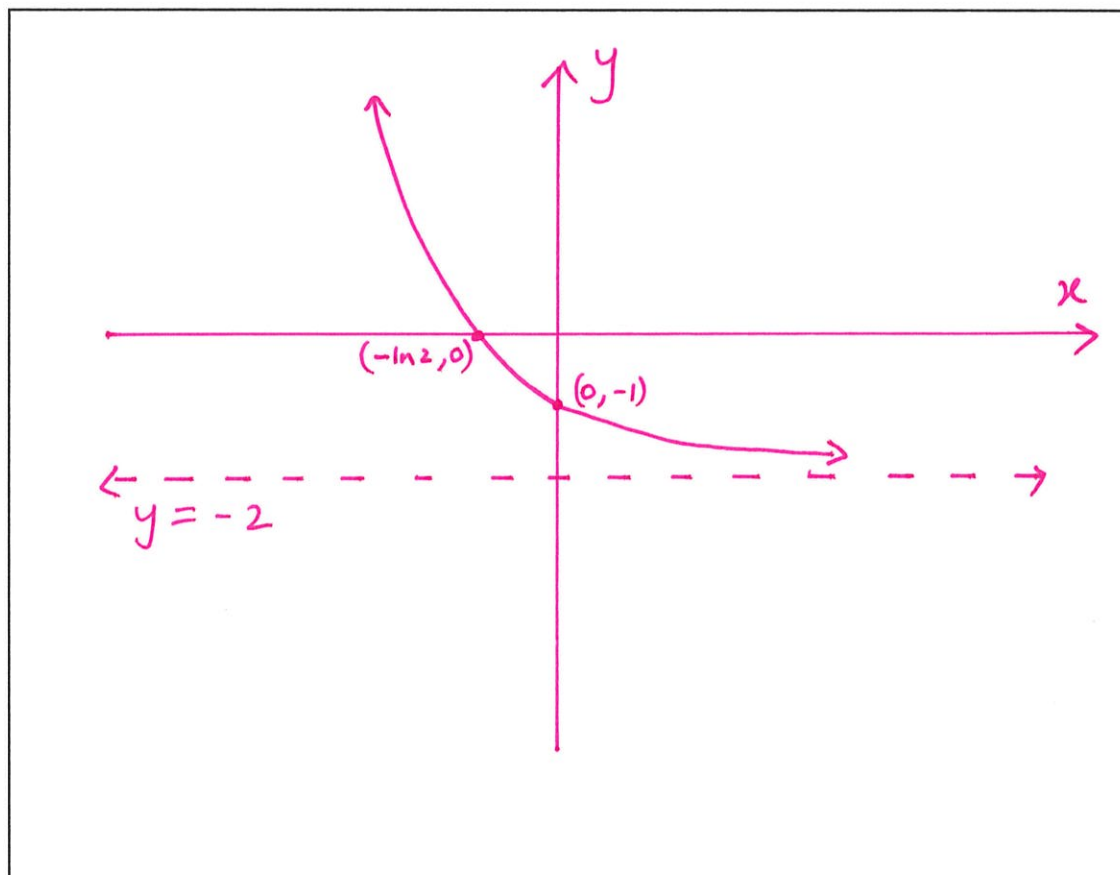
Question 7 (3 marks)

- (a) Sketch the graph of the following exponential function in the box below.
Label x and y intercepts and the asymptote.

3

$$y = e^{-x} - 2$$

Note: The box suggests the size of the expected graph



Asymptote: as $x \rightarrow \infty$, $y \rightarrow -2 \Rightarrow y = -2$ is the asymptote

① mark: correct asymptote and correctly drawn with a dotted line and labelled

x -int: set $y = 0$; $e^{-x} - 2 = 0$
 $e^{-x} = 2$

$$-x = \ln(2)$$

$$x = -\ln(2)$$

y -int: set $x = 0$: $y = e^0 - 2$
 $= -1$

① mark: correct intercepts ~~and~~ and correctly labelled

① mark: correct ~~y-int~~ and correctly labelled shape of curve

Question 8 (6 marks)

Sound intensity level (L) measured in decibels (dB) is a logarithmic measure given as:

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

The reference level of the intensity of sound (I_0) is 10^{-12} watts/m².

- (a) The table below shows the average sound intensity levels (L) and the intensity of the sound (I) of four instruments of a symphony orchestra.

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Cello	87	(i)
Violin	93	0.0019953
Trombone	100	0.0100000
Timpani and bass drum rolls	(ii)	0.0501187

Determine the missing values of (i) and (ii).

i) $87 = 10 \log_{10} \left(\frac{I}{I_0} \right)$

$\log_{10} \left(\frac{I}{I_0} \right) = \frac{87}{10}$

$\frac{I}{I_0} = 10^{8.7}$

$I = 10^{8.7} \times 10^{-12}$

$= 0.0005012$

correct working and

① mark: correct to 7 decimals

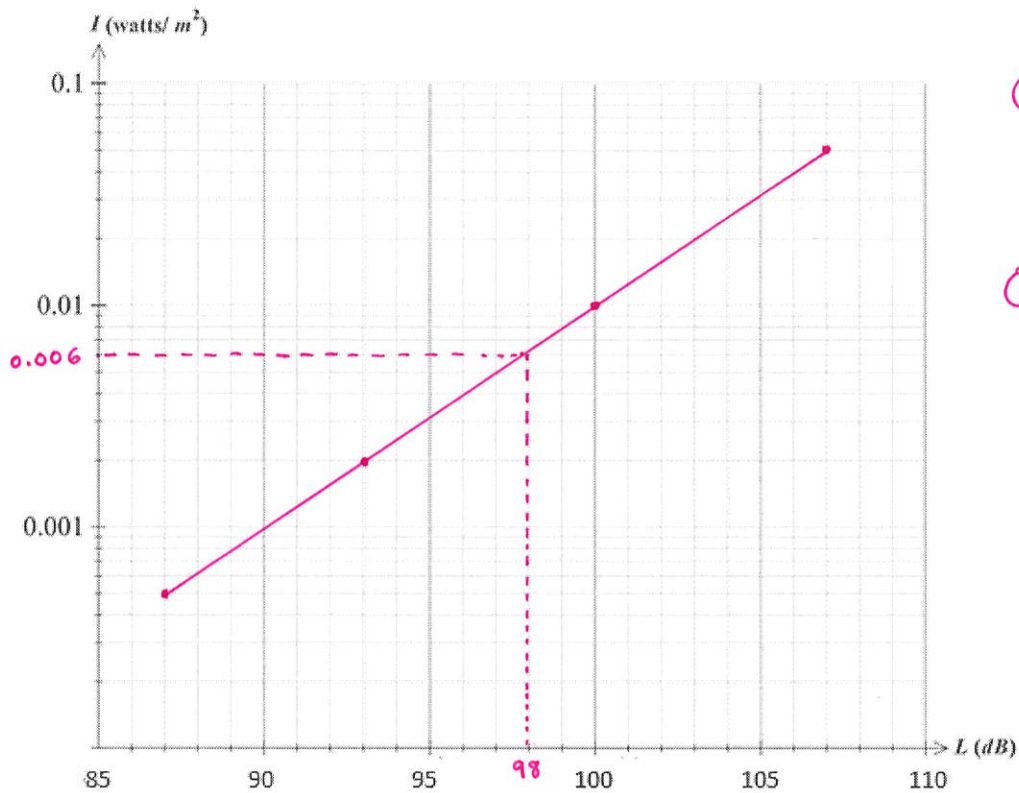
ii) $L = 10 \log_{10} \left(\frac{0.0501187}{10^{-12}} \right)$

$= 107$

① mark: correct to nearest dB

Examination continues on next page

- (b) Plot the intensity of sound (I) against the average sound intensity levels (L) for the **four instruments listed in part (a)** on the semi-logarithm graph below. 2
Connect the points with a line.



① mark:
correctly plotted
points

① mark:
lines drawn
through the
points

- (c) Using the graph determine the average sound intensity level (dB) of the sound from a musical instrument that has an intensity of 0.006 watts/m². 1

98 dB

① mark: correct answer
with dotted lines
shown on graph

- (d) Explain why a semi-logarithm grid is useful for this data. 1

It makes it easier to read the values from the graph compared to a linear scale on both axes, especially for very small or large values.

① mark: equivalent logical and sensible explanation

Question 9 (2 marks)

Given $\log_c(y) = \log_c(A) + tk \log_c(b)$, where A, b, c, k are (constant) positive real values with $b \neq 1$ and $c \neq 1$, and t and y are variables, show that $y = Ab^{kt}$.

2

$$\log_c(y) = \log_c(A) + \log_c(b^{kt})$$

$$\log_c(y) = \log_c(Ab^{kt})$$

① mark: correctly applies log laws

By taking exponential of both sides with base $c \Rightarrow y = Ab^{kt}$, as req.

① mark: correct answer with logical working

Examination continues on next page